14. Wednesday, April 30. More on induction (Chapter 4 part B)

Last Time: Induction. To prove $P(n)$ is true for all $n \in \mathbb{N}$,

1) Show $P(1)$ is true

2) Show that, for any $k \in \mathbb{N}$, $P(k) \rightarrow P(k+1)$ is true.

Theorem 1: For all natural numbers n, $\sum_{i=1}^{n}(2i-1) = n^2$.

Proof: For each $n \in \mathbb{N}$, let $P(n)$ be the statement $\sum_{i=1}^{n}(2i-1) = n^2$

We prove by induction on n that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** $P(1)$ says $\sum_{i=1}^{1}(2i-1) = 1^2$. But both sides are equal to 1, so $P(1)$ holds.
- **Inductive step:** Fix $k \in \mathbb{N}$. Want to prove $P(k) \rightarrow P(k+1)$

Assume $P(k)$ is true, so $\sum_{i=1}^{k}(2i-1) = k^2$.

Now, we want to show $P(k+1)$ is true, which means $\sum_{i=1}^{k+1}(2i-1) = (k+1)^2$. Compute

$$\sum_{i=1}^{k+1}(2i-1) = \sum_{i=1}^{k}(2i-1) + 2(k+1)-1 \underset{\text{IH}}{=} k^2 + 2k + 1 = (k+1)^2, \quad \text{as desired.}$$

Then, by induction, it follows that $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Technique ("Extended Induction" in your textbook, I'll just call this "Induction")

To prove that $P(n)$ is true for all $n \in \mathbb{N}$ with $n \geq N$

1) Base case: Show $P(N)$ is true

2) Inductive step: Show that for any $K \geq N$, $P(K) \rightarrow P(K+1)$ is true.

Example: The McNugget problem.

Suppose that at McDonald's you can order McNuggets in 4-piece or 5-piece orders.

Of course we can now order 4, 5, 8, 9, 10 McNuggets. Can't order 6.

What are the possibilities for McNuggets you can order?

1	No
2	No
3	No
4	4
5	5
6	No
7	No
8	4+4
9	4+5
10	5+5

11	No	
12	4+4+4	
13	4+4+5	} 4 in a row
14	4+5+5	} 4 in a row
15	5+5+5	} Three 5's → four 4's
16	4+4+4+4	
17	4+4+4+5	
18	4+4+5+5	

Theorem (McNugget Thm): For any $n \in \mathbb{N}$ with $n \geq 12$, it's possible to order exactly n McNuggets in batches of 4 or 5.

Proof: For $n \geq 12$, let $P(n)$ be the statement "Can order n"

Base case: $P(12)$ is true: We can order 12 nuggets by ordering three 4-piece servings.

Inductive step: Assume for some fixed $K \geq 12$, it is possible to order K nuggets

Want to show we can also order $K+1$ Nuggets. There are two cases for the K-piece order

Case 1: The K-piece order contains a 4-piece batch.

In this case, replace the 4-piece with a 5-piece batch to get $K+1$ Nuggets.

Case 2: The K-piece order contains no 4-piece batch.

In this case, we used only 5-piece batches. But since $K \geq 12$, we ordered at least three 5-pieces. Replace those three with four 4-pieces to get $K+1$.

Hence, by induction, we can order n Nuggets for any $n \geq 12$. □

Next: Subtleties

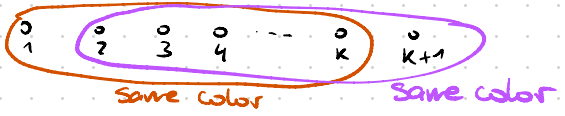
Non-Theorem: Let $n \in \mathbb{N}$. Then, in any collection of n horses, all of the horses are the same color

$P(n)$

"**Proof**": We proceed by induction on n.

Base case: For $n=1$, it's clear that in a collection of 1 horse all are the same color.

Inductive step: Assume the statement is true for some $k \in \mathbb{N}$. Let's show it's also true for $k+1$. Suppose we have a collection of $k+1$ horses, number them $1 - (k+1)$



Then, by ind hyp. in the collection $1-k$ all horses are the same color

in the collection $2-(k+1)$ all horses are the same color

Therefore horses $1, 2, \ldots, k+1$ are the same color

By induction, it follows that the statement is true for any $n \in \mathbb{N}$ □

What went wrong? In the inductive step, our argument does not work when $k=1$

For example, if we have $k+1=2$ horses of different colors, then the first $k=1$ are the same color and the last $k=1$ also, but can't deduce they are the same color